

Yan Yan

SR. TECH-LEAD MANAGER · AI RESEARCH SCIENTIST · LLM & RECOMMENDATION SYSTEMS

San Jose, California, USA

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SUMMARY

AI researcher and engineering leader with 14 years building large-scale ranking, retrieval, and recommendation systems at Meta, TikTok, LinkedIn, JD.com, and Yahoo. Currently focused on LLM-powered recommendation (LLM4Rec) and generative retrieval – shipped semantic-ID and multi-modal embedding based generative retrieval on Instagram Reels, delivering **+0.3% time spent** and multiple tier-1 metric wins. Track record of defining technical strategy, building high-performing teams (up to 30 engineers), and delivering business-critical results – from launching TikTok's \$1.4M/day ads vertical to driving IG Reels cold start and creator growth at Meta. Published 10+ papers at top venues (ICML, KDD, SIGIR) with 7 US patents and applications.

WORK EXPERIENCE

∞ Meta

Menlo Park, USA

SR. TECH-LEAD MANAGER / AI RESEARCH SCIENTIST, IG REELS — CONTENT COLD START & ECOSYSTEM INTELLIGENCE

Mar. 2025 - Present

- **Generative Retrieval (Semantic ID & Multi-Modal):** Built Reels' first production generative retrieval stack for content cold start, then drove two successive generations — a **semantic-ID** retriever that generates content identifiers directly, and a **multi-modal embedding** retriever grounded in video, audio, and text representations. Both shipped to production, together delivering **+0.3% time spent** and additional IG tier-1 metric wins.
- **LLM4Rec & Content Understanding:** Drove LLM-based content understanding for semantic analysis and early-stage quality prediction, enabling real-time candidate generation for fresh content carrying no prior engagement signal — the core unlock for cold-start distribution.
- **Creation Modeling (pCREATE / pINSPIRE):** Led creation-side modeling for the IG creator ecosystem, building pCREATE and pINSPIRE predictors over multiple distinct user journeys into content creation. Exceeded daily new Reels creation inspiration (HIPC) targets by 3x.
- **Cold Start Strategy & Creator Growth:** Owned the multi-half IG Reels cold start roadmap with Ranking, Integrity, and Star Search; aligned 4 partner teams on content pool promotion and quality gating. Significantly improved distribution for small creators, directly strengthening creator ecosystem health.

🎵 TikTok / ByteDance

San Jose, USA

ENGINEERING MANAGER, SHOP ADS TEAM (10 ENGINEERS)

Sept. 2021 - Mar. 2025

- **0→1 Ads Vertical:** Founded and scaled TikTok Shop's product listing ads from zero to **\$1.4M daily revenue** globally — the company's first in-app commerce ads product. Achieved 10%+ revenue take rate in SEA/US and 40%+ GMV penetration.
- **Catalog Listing Ads:** Launched open-loop catalog ads in TikTok feeds, aggregating products from multiple advertisers into a single ad unit. Reached \$50K USD daily revenue run rate.
- **Team Building:** Built the Shop Ads ranking team from 3 to 10 engineers across US and China. Promoted 2 engineers to senior and mentored 1 tech lead.

∞ Facebook / Meta

Menlo Park, USA

TECH LEAD, FORMAT ADS RANKING TEAM

Dec. 2018 - Aug. 2021

- **Ads Autoflow (VP-Sponsored):** Led Facebook's largest auto-format product series, building image filtering, video auto-cropping, and comments re-ranking. Delivered **3% total app revenue uplift** — one of the highest-impact individual projects in FB Ads that year.
- **Dynamic Creative Optimization:** Designed multi-field text optimization for DCO, driving 1%+ total revenue growth. Mentored 2 engineers to promotion.

🛒 JD.com

Beijing, CHN & Mountain View, USA

SR. MANAGER, ADS RECOMMENDATION TEAM (30 ENGINEERS)

Mar. 2017 - Dec. 2018

- **Ranking & Revenue:** Overhauled ranking model architecture and pricing strategy, lifting CTR 7% and ad revenue 10%. Built AutoML framework and productionized contextual bandits for 10%+ GMV increase.
- **Organization:** Managed 30-person ads recommendation team across Beijing and Mountain View. Defined quarterly roadmaps and led hiring for 12 positions.

- Built and led LinkedIn's 2016–2017 flagship Recruiter Search products (Search by Job, Search by Ideal Candidate), powering 40%+ of company revenue. Delivered 3%+ CTR lift and saved tens of millions via infrastructure migration.

- Built R-streaming traffic forecasting (<5% MAPE) for Yahoo's main properties and designed ads smart pricing algorithms for mobile search.

TECHNICAL SKILLS

- **Languages & Tools:** Python, Java, C++, SQL, AI-Assisted Development (Claude Code, Cursor, Copilot)
- **LLM & AI:** LLM/GenAI (Llama, Qwen, Gemini), vLLM, Generative Retrieval, Semantic IDs, Multi-Modal Embeddings, Content Understanding, RAG, Prompt Engineering
- **ML:** Deep Learning (PyTorch, TensorFlow), Large-Scale Ranking Models, Sequence Modeling, Online Learning, Contextual Bandits, AutoML, NLP
- **Ads & Recommendation:** LLM4Rec, Retrieval & Candidate Generation, Content Cold Start, Ads Ranking & Pricing, CTR/CVR Prediction, Dynamic Creative Optimization, Embedding Models
- **Infrastructure:** Distributed Systems, Real-Time Serving, A/B Testing, Data Pipelines, Multi-Objective Optimization
- **Leadership:** Engineering Management (up to 30 engineers), Tech Lead, Cross-Functional Product Strategy, Mentorship

EDUCATION

EECS Department, Northeastern University

PH.D OF SCIENCE

Boston, USA

Sept. 2006 - Oct. 2012

- Thesis: Learning from Imperfect and Related Labels

EECS Department, Tsinghua University

BACHELOR OF SCIENCE

Beijing, China

Sept. 2002 - Jun. 2006

SELECTED PUBLICATIONS & PATENTS

SELECTED PUBLICATIONS (10+ TOTAL; ICML, KDD, SIGIR, AISTATS, ECML, CIKM)

- Active Learning from Crowds. Yan Yan, Glenn M Fung, Rómer Rosales, Jennifer G Dy. **ICML**, 2011
- Medical Coding Classification by Leveraging Inter-Code Relationships. Yan Yan, Glenn Fung, Jennifer G Dy, Romer Rosales. **KDD**, 2010
- A Flexible Forecasting Framework for Hierarchical Time Series with Seasonal Patterns. Zitao Liu, Yan Yan, Milos Hauskrecht. **SIGIR**, 2018
- A Practical Deep Online Ranking System in E-Commerce Recommendation. Yan Yan, Zitao Liu, Meng Zhao, et al. **ECML**, 2018
- From Query-By-Keyword to Query-By-Example: LinkedIn Talent Search Approach. Viet Ha-Thuc, Yan Yan, et al. **CIKM**, 2017

PATENTS (7 US PATENTS & APPLICATIONS IN SEARCH RANKING AND RECOMMENDATION SYSTEMS)

- Ideal Candidate Search Ranking. US Patent App. 15/168,903
- Query Building for Search by Ideal Candidates. US Patent App. 15/168,811
- Automatic Labeler Assignment Using A Model Built from Multi-Labeler Data. US Patent 8,027,939